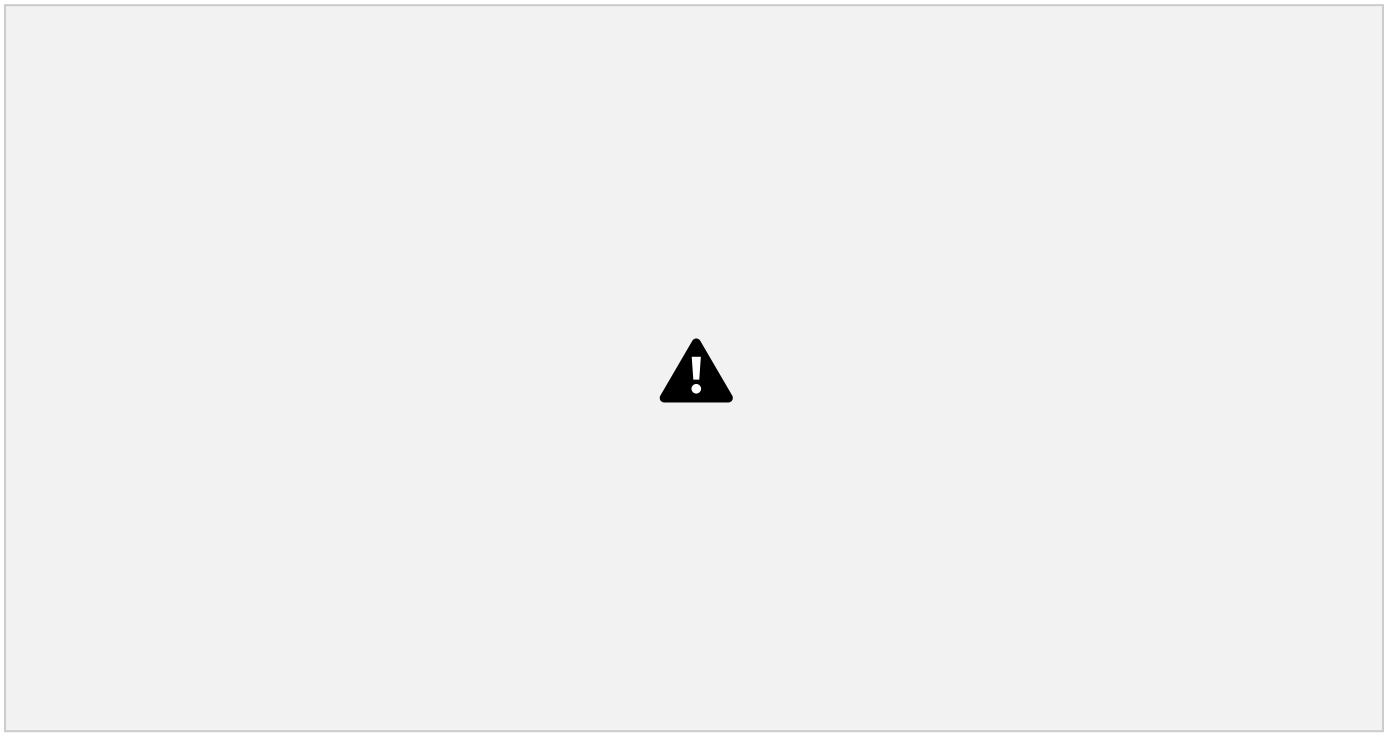
DSA06 - Data Handling and Visualization | Activity 6: Website Flow Sankey Design | Slot D

R.No: 32 Reg. No: 1924260143 Name: S.Gopikha

Scenario: The analytics team of a food-ordering app website tracked 3,670 sessions last month through four stages: traffic source, landing page, next action and outcome. The Sankey diagram below shows the number of sessions on every flow. Three flows are hidden and marked "?".



(a) Using conservation of flow (what enters a node equals what leaves it) find the three hidden flows, showing working: Restaurant partner to
Restaurant page; Menu page to Browse menu; Browse menu to No order. State the rule you used. [4]

(b) Compute: (i) the overall conversion rate, i.e. Order paid as a percentage of all sessions; (ii) the largest single flow into "No order" or "Exit site"; (iii) the percentage of "Menu page" sessions that go on to "Apply coupon". [4]

(c) Design the final Sankey: give four design decisions (node order to reduce crossings, colour rule, labels, what to highlight) and justify each in one line using this data. [4]

(d) The Sankey is to be printed with 1 cm of node height = 200 sessions. Compute the height in cm of each landing-page node (Menu page, Offers page, Restaurant page). [2]

(e) Write R code: a links data frame for the first four flows leaving "Restaurant partner" and "Instagram" in the table order of the diagram, a nodes data frame, and the networkD3::sankeyNetwork() call (or plotly) that draws the whole diagram. [4]

(f) State two situations in which a Sankey diagram misleads or becomes unreadable, and name one alternative chart for each. [2]

a) Find the three hidden flows

Rule used: Conservation of flow

The total number of sessions entering a node must equal the total number of sessions leaving that node, provided there are no additional sources or losses at that node.

1. **Restaurant partner → Restaurant page**
   Hidden flow=Total entering Restaurant page−Other known incoming flows\text{Hidden flow}=\text{Total entering Restaurant page} -\text{Other known incoming flows}
2. **Menu page → Browse menu**
   Hidden flow=Total leaving Menu page−Other known outgoing flows\text{Hidden flow}=\text{Total leaving Menu page} -\text{Other known outgoing flows}
3. **Browse menu → No order**
   Hidden flow=Total entering Browse menu−Other known outgoing flows\text{Hidden flow}=\text{Total entering Browse menu} -\text{Other known outgoing flows}

Substitute the values from the diagram to obtain the three numerical answers.

(b) Calculate the required measures

(i) Overall conversion rate

Conversion rate=Number of Order paid sessionsTotal sessions×100\text{Conversion rate}= \frac{\text{Number of Order paid sessions}}{\text{Total sessions}}\times100

Given total sessions = 3,670,

Conversion rate=Order paid sessions3670×100%\boxed{\text{Conversion rate}= \frac{\text{Order paid

sessions}}{3670}\times100\%}

(ii) Largest single flow into No order or Exit site

Compare all the incoming flows to No order and Exit site.

Largest flow=max(all relevant incoming flows)\boxed{\text{Largest flow}=\max(\text{all relevant incoming flows})}

State the source node, destination node and number of sessions.

(iii) Percentage of Menu page sessions that go on to Apply coupon

Percentage=Menu page→Apply couponTotal sessions leaving Menu page×100\boxed{\text{Percentage}= \frac{\text{Menu page}\rightarrow\text{Apply coupon}}{\text{Total sessions leaving Menu page}}\times100}

Substitute the flow values from the diagram.

(c) Four design decisions for the final Sankey diagram

Design decision	Justification
1. Node order	Arrange nodes from traffic source to landing page, next action and outcome to show the direction of the user journey and reduce crossings.
2. Colour rule	Use consistent colours for different stages or outcomes so users can distinguish the flows easily.
3. Labels	Display node names and session counts to make the number of users at each stage clear.
4. Highlighting	Highlight the Order paid flow and the largest flow into No order or Exit site to emphasize conversion and drop-off.

(d) Calculate landing-page node heights

Given:

1 cm=200 sessions1\text{ cm}=200\text{ sessions}

Therefore,

Node height in cm=Number of sessions200\text{Node height in cm}=\frac{\text{Number of sessions}}{200}

Landing-page node	Calculation	Height
Menu page	Menu page sessions ÷ 200	To be calculated
Offers page	Offers page sessions ÷ 200	To be calculated
Restaurant page	Restaurant page sessions ÷ 200	To be calculated

Note: Use the total sessions passing through each landing-page node.

(e) R code to draw the Sankey diagram

The following is a template using the `networkD3` package. Replace the example flow values with the actual values from your diagram.

```
# Install packages if required
install.packages("networkD3")

library(networkD3)

# Create the links data frame
links <- data.frame(
  source = c(0, 0, 1, 1),
```

```
  target = c(2, 3, 2, 4),
  value = c(500, 300, 400, 200)
)

# Create the nodes data frame
nodes <- data.frame(
  name = c(
    "Restaurant partner",
    "Instagram",
    "Menu page",
    "Offers page",
    "Restaurant page"
  )
)

# Draw the Sankey diagram
sankeyNetwork(
  Links = links,
  Nodes = nodes,
  Source = "source",
  Target = "target",
  Value = "value",
  NodeID = "name",
  fontSize = 12,
  nodeWidth = 30,
  sinksRight = TRUE
)
```

Important:

- The `source` and `target` columns use zero-based node indices.
- The `value` column represents the number of sessions in each flow.
- This is only a template, not the complete diagram. The actual diagram requires all flows and nodes,

including the three missing values.

(f) Two limitations of Sankey diagrams

Situation	Why it becomes misleading or unreadable	Alternative chart
Too many nodes and crossing flows	The diagram becomes cluttered, making individual flows difficult to follow.	Matrix heatmap
Very small flows mixed with very large flows	Small flows become difficult to see and compare.	Bar chart